

Exercise 6.2.1 - Convert into first-order form

$$\frac{d^2x}{dt^2} - x \frac{dx}{dt} + x^2 = 0, \quad y = \frac{dx}{dt} \quad (1)$$

$$\frac{dy}{dt} - xy + x^2 = 0 \quad (2)$$

$$\frac{dy}{dt} = xy - x^2 \quad (3)$$

Exercise 6.2.2 - Convert into autonomous first-order form

$$\frac{d^2x}{dt^2} - a \cos(bt) = 0, \quad \frac{d\tau}{dt} = 1, \quad y = \frac{dx}{dt} \quad (4)$$

$$\frac{dy}{dt} - a \cos(b\tau) = 0, \quad \tau(0) = 0, \quad \tau(t) = t \quad (5)$$